
UNIT 12: APPLICATIONS OF AI

Structure	Page No.
12.0 Introduction	372
12.1 Objective	373
12.2 AI in healthcare	373
12.3 AI in finance	376
12.4 AI in transportation	379
12.5 AI in customer service and chatbots	382
12.6 AI in education	384
12.7 Summary	386
12.8 Solutions/ Answers	387
12.9 Further Readings	389

12.0 INTRODUCTION

Artificial Intelligence (AI) has evolved into an advanced field of computer science. It significantly transforms multiple domains by enabling machines to think, learn, and act intelligently. With the rapid growth of digital systems, organizations across various domains are generating a huge amount of data. However, traditional computational approaches face challenges in extracting meaningful insights, making timely decisions, and identifying complex patterns from such large-scale data. These limitations highlight the need for intelligent systems that are capable of learning from data and making decisions in dynamic environments.

In healthcare, the increasing complexity of medical data, such as imaging records, electronic health reports, and patient histories, poses significant challenges for accurate diagnosis and timely intervention. Manual processing is time-consuming and may lead to variability, thereby affecting clinical outcomes. Moreover, the growing demand for personalized treatment and early disease prediction places additional pressure on existing healthcare systems.

Similarly, the education sector faces challenges in addressing diverse learning needs, monitoring individual student progress, and ensuring fair and consistent evaluation. Traditional instructional models often struggle to adapt to varying learner capabilities, while educators are burdened with repetitive assessment and administrative tasks, limiting their ability to focus on personalized mentoring.

In finance, increased digital transactions and financial services have exposed financial institutions to risks such as fraud, market volatility, and credit default. Identifying suspicious patterns, managing financial risk, and delivering efficient customer support in real time remain critical concerns for financial institutions operating in highly competitive and regulated environments.

Transportation and industrial systems are also becoming increasingly complex due to urbanization, growing logistics demands, and the need for operational

efficiency. Challenges such as traffic congestion, unpredictable delays, equipment failures, and inefficient resource utilization highlight the limitations of rule-based and manual control mechanisms. Similar concerns arise in manufacturing, where maintaining productivity while minimizing downtime and defects is a persistent issue.

Agriculture faces its own set of challenges, including climate variability, resource constraints, pest management, and the need to improve crop yield sustainably. Traditional farming practices often lack the precision and adaptability required to respond effectively to these evolving conditions.

Beyond these domains, areas such as cybersecurity, environmental monitoring, entertainment, and space exploration are confronted with increasingly complex problem spaces that require systems capable of reasoning under uncertainty and adapting over time. Addressing these challenges necessitates intelligent computational approaches that can learn from experience and support complex decision-making.

12.1 OBJECTIVES

After going through this unit, you will be able to:

1. Understand the importance of Artificial Intelligence (AI) for solving real-world problems
2. Discuss the use of AI in various domains such as healthcare, finance, transportation, customer services and education.

12.2 AI IN HEALTHCARE

Artificial Intelligence (AI) has become one of the important decision-supporting tools in the healthcare sector. With the use of AI, the health sector has become faster, more accurate, and more accessible. AI-based systems can assist doctors in identifying disease stages and personalising the treatment. Unlike traditional methods, AI systems can analyze massive amounts of data in seconds and identify patterns that even highly trained human experts may miss. It can be applied across diagnosis, treatment planning, drug discovery, and medical imaging.

The healthcare system faces lots of challenges worldwide, such as a shortage of doctors, increasing patient demand, and rising healthcare costs. AI acts as a digital assistant for doctors and helps them speed up the work process. This may help in improving patient outcomes and reducing human errors. According to a report by Accenture in 2022, AI applications in healthcare can potentially create \$150 billion in annual savings for the U.S. healthcare system by 2026. Also, NITI Aayog and other initiatives estimate that AI could add hundreds of billions of dollars to India's economy by the mid-2030s, reflecting broad adoption, including in healthcare.

12.2.1 AI in Diagnosis

Diagnosis is one of the most critical stages in healthcare. An early and accurate diagnosis often determines the success of treatment. AI can assist the medical practitioners in diagnosing diseases by analysing electronic health records (EHRs), lab test results, genetic data, and even patient-reported symptoms. Machine learning models can be trained on thousands of past cases to detect signs of diseases such as cancer, diabetes, Alzheimer's, cardiovascular diseases, and rare disorders.

For example, IBM Watson Health uses AI-based systems to analyze patient records and compare symptoms with a vast amount of medical literature to suggest possible conditions. Similarly, AI chatbots such as Babylon Health allow patients to input their symptoms and receive possible diagnoses before visiting a doctor.

Some of the applications of AI-based diagnosis are:

- AI algorithms are widely used in oncology. For example, Google's LYNA (Lymph Node Assistant) can detect metastatic breast cancer with an accuracy of over 92%. Further, AI can also scan through various medical imaging such as mammograms, lung CT scans, and skin images to identify cancerous cells at an early stage.
- AI helps in cardiology by analyzing ECG (Electrocardiogram) and echocardiogram signals to detect heart-related issues.
- AI models are very helpful in Neurology. It can identify early signs of Alzheimer's or Parkinson's by analyzing brain scans and patient behavior data.
- AI can be used to assess an individual's mental health using questionnaires, textual data, facial expressions, voice, etc.

12.2.2 AI in Treatment

AI plays an important role in treatment planning and execution, once diagnosis is done. Some of its applications are as follows:

1. Using AI, precision or personalised treatment can be provided to patients, which means customizing the treatments to individual patients rather than using a "one-size-fits-all" approach. For example, in cancer treatment, AI can analyze a patient's genetic profile to recommend the most effective drug.
2. AI-based robots can act as an assistant to surgeons in performing minimally invasive surgeries with high precision. These systems can analyze real-time surgical data and alert doctors if there's a risk.
3. Developing a new drug is an expensive and time-consuming task. AI helps accelerate the process by predicting how different molecules will interact with diseases. For instance, the company DeepMind

(Alphabet/Google) developed AlphaFold, which can predict protein structures with remarkable accuracy.

4. AI also helps in tracking long-term diseases such as diabetes and hypertension. AI-enabled wearable devices can continuously monitor patient data and generate alerts. For example, Apple Watch and Fitbit use AI algorithms to detect abnormal heart rhythms and warn users.

12.2.3. AI in Medical Imaging

Medical imaging is one of the most advanced and successful applications of AI in healthcare. AI-based computer vision models analyse X-rays, MRIs, CT scans, and ultrasounds to detect abnormalities.

- AI-based systems can assist radiologists by automatically identifying the suspected areas in the scan. For example, AI can help detect the lung nodules in the CT scans that may be an indication of early-stage lung cancer.
- Digital pathology uses AI to analyze the biopsy samples. Instead of manually examining tissue slides, AI models can scan and classify the tissue patterns and give us results faster.
- AI systems like Google's DeepMind AI can detect diabetic retinopathy and macular degeneration from retinal images.
- The AI tools are trained to analyze the skin images to identify melanoma and other skin cancers. Apps such as SkinVision allow patients to upload pictures of skin lesions for AI analysis.

12.2.4. Case Study: IBM Watson for Oncology

IBM Watson for Oncology [1] is one of the most well-known AI applications in the healthcare sector.. It is developed by IBM in collaboration with Memorial Sloan Kettering Cancer Center (MSKCC). This AI system was designed to help doctors in cancer diagnosis and the recommendation of personalized treatment plans. The main goal of the system is to use the vast amount of data available in the medical literature, clinical guidelines, and patient records to help oncologists make fast and precise decisions.

Watson for Oncology uses the latest technologies such as natural language processing (NLP) and machine learning algorithms to read and understand the unstructured medical data such as clinical notes, pathology reports, and research papers. It processes millions of pages of medical information and then compares this knowledge to patient's medical history, test results, and genetic data. Based on the analysis of data, Watson model suggests the potential treatment options which are ranked in three classes as "recommended," "for consideration," or "not recommended," along with the scientific evidence to support each option.

This AI system is in use at several hospitals across many countries, including the United States, India, and South Korea. For example, Manipal Hospitals in India implemented Watson for Oncology to support cancer specialists in

identifying the optimal treatment plans for the patients. Using the system, less time is spent reviewing cases, and more precise treatment can be provided to patients.

Despite its wide use in healthcare, Watson for Oncology faced challenges of adapting to local treatment protocols and rapidly evolving medical data.

12.2.5. Challenges and Ethical Concerns

The ethical concerns and challenges faced by AI-based healthcare systems are:

- Medical data is highly sensitive. It can harm patients, if misused.
- If an AI-based system is trained on non-representative datasets, it may lead to incorrect results, especially in underrepresented populations.
- AI-based healthcare systems can be used to assist doctors. They cannot replace doctors. The final decisions must remain with human experts.
- Who is responsible if an AI system makes a wrong diagnosis?

12.3 AI IN FINANCE

Finance is one of the most data-driven sectors in the world. Every second, banks, stock markets, and insurance companies generate enormous amounts of data. Analyzing this huge amount of data (also called big data) fast and accurately is important for making business decisions. To make timely decisions after due analysis of data, AI based systems can play an important role. They can help to protect us from fraud, and ensure the financial stability.

In the finance sector, AI technologies such as machine learning (ML), deep learning (DL), and natural language processing (NLP) are helping to improve the accuracy, efficiency, and security. It is helping to prevent credit card fraud, executing the stock trades in milliseconds and predicting the risks in the loans and investments.

12.3.1. AI in Fraud Detection

One of the biggest challenges in the financial industry is digital financial fraud. The traditional methods for fraud detection are often based on a fixed set of rules e.g., if a transaction is above a certain amount, generate an alert. However, the fraudsters are constantly evolving their techniques and making the rule-based systems less effective.

AI-based systems use machine learning models to continuously learn from data generated by new transactions and adapt to changing patterns in fraud. These models can detect unusual patterns or anomalies in real time.

Some of the real-world examples in fraud detection are:

1. AI systems can monitor millions of transactions and flag suspicious activity, e.g., if a card is used in two countries within minutes.
2. AI can analyse login attempts and detect abnormal behaviour, such as accessing an account from an unusual device.

3. AI models can scan claims and medical records to detect false claims in insurance.

12.3.2. AI in Algorithmic Trading

Stock trading is a highly competitive and time-sensitive task. Timeliness of the decisions can make a huge difference in outcomes. It can result in significant profits or losses. It is difficult for human traders to analyze market data at very high speed.

Algorithmic trading (or algo-trading) refers to the use of computer programs to execute trades based on pre-defined rules. In these algorithms, AI can be used to learn patterns, predict price movements, and optimize the strategies in real time as follows:

- AI models can analyze historical stock prices, market news, and economic indicators to predict future price movements.
- NLP models can read financial news, tweets, and company reports to identify market sentiment and predict stock fluctuations.
- AI systems can execute thousands of trades per second by rapidly analyzing real-time market data.
- AI algorithms can help investors to balance risk and return by suggesting the best combinations of assets.

Examples of real-world implementation are:

1. JP Morgan's LOXM is an AI-based system which can execute trades with minimal market impact.
2. Goldman Sachs uses AI-based platforms to analyze the investment opportunities.
3. Renaissance Technologies, a hedge fund, relies on AI and ML models for its trading strategies..

12.3.3. AI in Risk Assessment

While granting a loan, insuring a customer, or investing in the stock market, every financial institution should assess the risks. Traditional risk assessment methods are based on credit scores and static rules. AI-based systems can find even the hidden patterns and perform an accurate assessment in different fields such as:

1. AI-based systems can evaluate whether a person is likely to repay a loan by analyzing not just the credit score but also their spending habits, digital footprint, and social behavior.
2. AI-based models can forecast the market volatility and the potential losses by analyzing the real-time financial data, global events, and economic reports.

3. AI-based systems can help banks to detect internal risks like system failures, cyberattacks, or staff misconduct.
4. AI-based models can assess risks in insurance applications. For instance, they can analyze the driving behavior from the telematics devices in the cars to determine fair insurance premiums.

Examples of real-world implementation are:

- ZestFinance uses AI models to analyze the alternative data (e.g., phone bill payments, online behavior) for better credit scoring.
- FICO incorporates AI into its credit scoring system to reduce defaults.
- Swiss Re (insurance company) uses AI to improve underwriting and predict claims risks.

12.3.4. Case Study: JPMorgan Chase

JPMorgan Chase's [2] COiN (Contract Intelligence) platform is one of the most successful and widely mentioned examples of artificial intelligence in the financial sector. The system was developed to automate the process of reviewing the legal and financial documents, which traditionally took thousands of employee hours every year. COiN can analyze the complex commercial loan agreements very fast using ML and NLP. It can accurately identify the important clauses, risks, and compliance requirements.

Manual review of loan documents is a time-consuming process that can be affected by human error as well. Using COiN, JPMorgan has reduced document review time from 360,000 hours to just a few seconds. This system not only saves time and cost but also ensures greater consistency and accuracy in legal and financial operations.

In addition to the contract analysis, JPMorgan Chase also applies AI-based models for fraud detection and transaction monitoring. AI algorithms continuously analyze customer transactions in real-time and identify unusual patterns that may indicate fraudulent activity. These models learn new patterns over time and improve their ability to detect new types of financial fraud that can be missed by traditional rule-based systems. This approach improves customer trust, reduces financial risk, and sets a benchmark for the use of AI in global finance.

Check your Progress 1

1. What are the challenges of using AI in healthcare?

.....

.....

.....

2. How AI can ensure security in the domain of finance?

-
-
-
3. How does bias in medical training datasets affect AI-driven clinical decisions? Suggest possible solutions.
-
-
-

12.4 AI IN TRANSPORTATION

Transportation is one of the most important sectors for society and the economy. Transportation is important as people need to travel for work, education, health care, and leisure. Further, goods need to be transferred from producers to consumers. However, traditional transportation systems impose a number of challenges such as traffic congestion, road accidents, high fuel consumption, and environmental pollution. To cater to these challenges, AI has evolved as a powerful tool. Using machine learning, computer vision, robotics, and big data, AI systems can improve vehicle operations, traffic flows, and transport networks.

Two major applications of AI in transportation are:

1. Autonomous vehicles such as self-driving cars, trucks, and drones.
2. Traffic optimization such as intelligent traffic management, smart routing, and predictive analysis.

12.4.1. Autonomous Vehicles (Self-Driving Cars)

Autonomous vehicles (AVs), also known as self-driving vehicles, are intelligent transportation systems that operate without human intervention by using sensors, artificial intelligence, machine learning, and advanced control systems to perceive their environment, make real-time decisions, and navigate safely with minimal or no driver input under defined operating conditions.

Autonomous vehicles are based on many technologies, such as:

- Cameras, LiDAR (Light Detection and Ranging), radar, and ultrasonic sensors collect information about a vehicle's surroundings.
- AI-based models process sensor data to identify objects such as cars, pedestrians, road signs, and obstacles.
- Vehicles use GPS and AI based algorithms to estimate their position on a map.
- Machine learning models help to decide actions like braking, accelerating, or changing lanes.

- The computer system of the car can map the decisions into movement through steering, throttle, and brakes.

The Society of Automotive Engineers (SAE) defines a six-level stack of driving automation, i.e. from Level 0 (no automation) to Level 5 (full automation without human input). Most current commercial systems like Tesla Autopilot or GM Cruise are developed around Level 2 or 3, which indicates that they require human intervention at some point in time.

Some of the real-world examples of the autonomous vehicles are:

- Google's self-driving project, Waymo, operates the autonomous taxis in cities like Phoenix, Arizona.
- Tesla's Autopilot uses computer vision and neural networks for semi-autonomous driving.
- Nuro focuses on autonomous delivery robots for groceries and small goods.
- Uber ATG previously developed self-driving taxis (sold to Aurora in 2020).

Some of the challenges of the autonomous vehicles are as follows:

- The safety and reliability concerns if an autonomous car gets involved in some accident.
- The LiDAR sensors and other high-performance AI based systems are very expensive.
- The roads, signals, and the communication networks need upgradation for full automation.
- Many people still hesitate to trust AI with their safety.

12.4.2. AI in Traffic Optimization

AI can be used to improve the existing transport systems by making them smarter. Cities around the world often struggle with traffic congestion, long commute times, increased fuel consumption, air pollution, greenhouse gas emissions and hence, economic losses. To overcome these problems, AI can be used for predicting traffic flow, optimizing signal timings, and guiding vehicles through the best routes.

Some of the applications for traffic optimization are:

- AI-based smart traffic lights can adjust signal timings based on real-time traffic flow. Instead of fixed timers, the signals can adapt it dynamically.
- Machine learning-based predictive analysis can help in predicting the traffic jams in advance by analyzing the historical data, weather and accidents.

- The GPS based smart navigation systems such as Google Maps and Waze use AI techniques to suggest the fastest routes.
- Using connected vehicles technology, the cars can communicate with each other which can help in cooperative driving.
- AI can help in bus scheduling, demand forecasting, and route planning to reduce waiting time.

Some of the real-world applications of AI-based traffic optimization are:

- Surtrac (Pittsburgh, USA) is an AI-based traffic light system that reduces travel time by 25% and wait time by 40%.
- Google Maps can predict traffic delays and suggest some alternative routes using machine learning.
- Singapore Smart Nation uses sensors and analytics for integrated traffic monitoring.
- China's Smart Cities uses AI for traffic signals optimization and emergency response.

12.4.3. Case Study: Tesla AI-Based Autonomous Vehicles

Tesla's self-driving car [3] is one of the widely known examples of artificial intelligence in transportation. AI-based projects such as Autopilot and Full Self-Driving (FSD) systems make the vehicles smarter so that they can navigate, steer, accelerate, and apply brakes automatically under various conditions. These systems are created using deep learning, computer vision, and sensor fusion, which allows the car to sense the surrounding and make decisions in real time.

Tesla vehicles are equipped with multiple cameras, ultrasonic sensors, and radar systems that can continuously capture environmental data. This data is processed using neural networks to identify the objects on the road, such as traffic lights, pedestrians, lane markings, other vehicles, and road signs. Unlike the traditional rule-based systems that depend on a fixed set of instructions, Tesla's autonomous vehicles learn from the massive amount of real-world driving data collected from millions of cars worldwide, which makes the model smarter.

The core idea behind Tesla's approach is the state-space search algorithms and decision-making, where the constant evaluation of all possible actions (such as turning, stopping, or accelerating) takes place to select the safest and most efficient option. This process happens within a short time and ensures the smooth and safe navigation even in complex traffic situations.

Tesla's AI-driven transportation technology has revolutionized the automotive industry, ranging from autonomous driving, traffic optimization to road safety improvements. It also supports the energy efficiency by optimizing the routes and the driving patterns. While fully autonomous driving is still under

development and faces regulatory challenges, Tesla's autonomous vehicles represent a major step toward the future of smart mobility, where vehicles can communicate, learn, and make decisions without human intervention.

12.5 AI in Customer Service and Chatbots

Across most industries, customers need support from service providers to answer their questions and listen to their complaints. Good customer service increases customer trust and satisfaction. Earlier, customer service was provided by human customer care executives who used to provide support on phone calls, emails, or in-person interactions. With the evolution of AI, a significant transformation has taken place in the customer service industry. One of the most significant innovations in this area is the use of chatbots and AI-based virtual assistants as alternatives to human service executives. These technologies allow companies to provide 24/7 customer support, reduce costs, and improve customer satisfaction.

12.5.1. AI based techniques to improve Customer Service

The following AI-based techniques are used to improve customer service:

1. Natural Language Processing (NLP) is used by chatbots to understand and communicate in human language.
2. Machine Learning can be used to analyse customer feedback and recommend more effective responses for future interactions.
3. Sentiment analysis can be used to detect the emotions of the customer i.e. whether a customer is angry, confused, or satisfied, and adjust responses accordingly.
4. AI can make personalized recommendation of products or services based on customer browsing history and preferences. This makes interactions more useful and relevant.

12.5.2. Applications of AI in Customer Service

Various applications of AI-based systems in Customer Service are:

- Unlike human agents, AI-based chatbots are available 24/7. They can handle customer requests around the clock, across different time zones.
- During peak hours, AI-based chatbots can manage thousands of queries simultaneously.
- Many banks, e-commerce platforms, and applications use AI-based chatbots to guide new customers for initial setup and registration process.
- Retailers use these chatbots to help customers to check delivery status, request refunds, or change orders.
- AI-based customer service may provide troubleshooting steps for the common problems before escalating it to a human executive.

- AI-based chatbots can provide support in multiple languages.

12.5.3. Case Study: Amazon Alexa

Alexa [4] is a voice-based AI assistant developed by Amazon. It is one of the most widely used AI-powered virtual assistants for customer support. It improves the customer interaction, generates automated responses, and provides a personalized assistance. It uses various latest technologies such as NLP, ML and speech recognition to understand spoken commands, process user intent, and respond naturally.

It allows users to perform a number of tasks such as checking product prices, placing orders, tracking deliveries, and managing subscriptions. All these tasks can be done simply using voice commands. Technically, Alexa uses rule-based state-space search to identify user intent. It then matches the identified intent with a suitable response and executes the required actions. Over time, its machine learning algorithms continuously learn from millions of interactions to improve accuracy and provide more personalized assistance.

By using Alexa as a virtual customer care executive, organizations can reduce their dependence on human call-centre agents. Some companies are integrating this virtual customer care executive with their customer service to provide 24/7 support. For instance, many smart home device manufacturers now use Alexa to help customers set up and control their products using conversational instructions. The advantages of using virtual customer care executive include faster response time, 24X7 availability, improved user satisfaction and less overheads.

Check your Progress 2

1. How do sensor fusion techniques (LiDAR, radar, cameras) improve decision-making in autonomous vehicles, and what limitations still exist?
.....
.....
.....
2. What is the role of sentiment analysis in the case of AI based customer services?
.....
.....
.....
3. Discuss the potential risks of relying heavily on AI for customer complaint resolution.

.....
.....
.....

12.6 AI IN EDUCATION

Education serves as the foundation of human progress. Traditional teaching methods often follow a one-size-fits-all approach, where all students are expected to learn the same content at the same pace. However, learners differ significantly in their abilities, preferences, and learning speeds. Some students grasp concepts more quickly than others, while some prefer visual explanations over textual content. Others require additional practice questions rather than advanced, challenging material.

AI has transformed the education system by enabling personalized learning experiences for individual students. AI-based Intelligent Tutoring Systems (ITS) function as virtual tutors, providing customized instruction tailored to each learner's needs. These technologies enhance learning effectiveness and also assist teachers in classroom management and monitoring student progress.

12.6.1. Personalized Learning with AI

- Personalized learning involves adapting teaching content, speed and learning style according to a student's needs. Instead of teaching everyone in the same way, AI system can adjust according to how well a student is performing. For example, if one student struggles with algebra, the AI system can provide additional exercises, step-by-step hints or video explanations on the specific topic to the student so that s/he can practice more on that topic. If another student finds the same topic easy, the system can give more advanced problems from the same topic to keep them challenged.
- Some of the features of AI-based personalized learning systems are:
- Khan Academy and DreamBox are using AI-based tools to adjust the difficulty of questions according to the student's needs in real time.
- These tools also present the learning analytics (test scores, speed, mistakes) of the student to find the patterns and identify the weak areas.
- These AI-based tools also recommend the lessons, exercises, or videos either based on a student's weak areas or in an incremental approach.
- AI-based learning games motivate the students by rewarding progress with points, badges, or levels.
- Some of the real-world examples of personalized learning are:
- Knewton (Pearson) adapts digital textbooks according to the individual student's needs.

- Coursera AI Recommendations can suggest courses to students based on their progress and interests.
- BYJU's (India) used AI for adaptive video lessons and quizzes.

12.6.2. Intelligent Tutoring Systems (ITS)

- An Intelligent Tutoring System (ITS) is a computer program that provides one-to-one tutoring to learners, similar to that offered by a human teacher. It uses AI to understand the student's response, provides hints, corrects mistakes, and offers feedback in real time.
- The components of ITS are:
- The domain model stores knowledge about the subject (e.g., math rules, grammar).
- The student model tracks the student's learning pace, weaknesses, and misconceptions.
- The tutoring model decides which lesson or hint should be provided to the student next.
- The user interface allows the student to interact with the system using text, voice, or visuals.

Some real-world examples of ITS are

- Carnegie Learning (Cognitive Tutor) is an AI-based tutor for mathematics.
- AutoTutor helps the students in science and technology subjects using natural language dialogue.
- Duolingo uses AI-based tutoring for personalized language learning.

12.6.3. Case Study: Duolingo

Duolingo [5] is a very well-known language learning platform that allows users to learn new languages. It uses AI to provide a personalized and adaptive learning experience to millions of users worldwide. It combines rule-based AI and machine learning to make learning engaging, effective, and customized according to each learner's ability.

It analyzes the pattern of users' interaction with given lessons and tasks to record the response time, accuracy, and mistakes. These recorded parameters are further used to predict the learner's proficiency level. Based on this data, the app dynamically adjusts the difficulty level of the exercises and the pace of new lessons. This approach is based on the state-space search concept where AI continuously evaluates the learner's current "state" and determines the optimal next "state" or lesson to maximize progress.

A rule-based system ensures that the grammar and the vocabulary rules are presented in some logical sequence. The machine learning algorithms improve the experience of personalization by identifying the weak topics for focused practice. Duolingo also uses NLP to evaluate pronunciation and the written answers to provide instant feedback similar to a human tutor.

The gamification feature of the app, such as streak, points, and leaderboard, is also based on AI algorithms. It maintains the user motivation by giving adaptive rewards. By combining psychology, linguistics, and artificial intelligence, Duolingo makes learning fun and scientifically effective.

This case study highlights the transformation in the field of education from a one-size-fits-all model to a personalized, interactive, and continuous learning experience using AI-based techniques. It shows that AI can act as a virtual tutor that is capable of understanding the learners' needs, adapting to their pace, and helping them to achieve better outcomes.

Check your Progress 3

1. Explain how AI-based adaptive learning systems improve student performance.

.....

.....

.....

2. How might over-reliance on AI tools affect students' critical thinking and problem-solving skills?

.....

.....

.....

3. How should teachers' roles evolve in classrooms increasingly supported by AI technologies?

.....

.....

.....

12.7 SUMMARY

This unit explores various applications and growing importance of Artificial Intelligence (AI) across a number of domains such as healthcare, finance, transportation, customer services and education. AI has become a transforming

force by enabling machines to process huge volume of data. It can learn patterns and experience from available data. It can perform complex tasks with high speed and improved accuracy.

In healthcare, AI provides support in medical imaging, disease prediction, drug development and personalized treatments with improved diagnosis accuracy. In financial sector, AI can help in improving security by fraud detection, automated trading, credit score estimation and improved customer services using intelligent chatbot.

Further, transportation system also has been revolutionized by AI based self-driving vehicles, intelligent traffic management and route optimizations which contributes to a safe and efficient mobility. In customer services, AI based chatbot and virtual assistant provides instant support round the clock which improves user satisfaction and reduces the operation cost. Similarly, in education AI provides adaptive learning system with personalized feedback and automated grading. It ensures that the teaching and learning process become more effective and student centric.

Across all these domains, in addition to enhanced efficiency, AI based models also provide advanced analytical capabilities and data-driven insights for effective decision making. However, it also brings some challenges such as data privacy concerns, ethical issues and need of human oversight. Overall, AI has become indispensable technology which is reshaping the industries and redefining the human interaction with machines. The unit emphasizes that the understanding of AI based applications and its implications is essential for preparing students and professionals to participate effectively in increasingly intelligent and automated world.

12.8 SOLUTIONS/ ANSWERS

Check your Progress 1

1. The challenges of using AI in healthcare include data privacy issues, lack of quality medical data, high implementation costs, and the need for expert supervision.
2. AI ensures security in finance by detecting fraud, monitoring transactions for unusual patterns, and protecting customer data using advanced algorithms.
3. Bias in medical training datasets can lead to inaccurate diagnoses and unequal treatment recommendations, particularly for underrepresented populations. This may worsen healthcare disparities and reduce trust in AI systems. Solutions include using diverse and representative datasets, continuous bias auditing, transparent algorithms, inclusive clinical validation, and involving multidisciplinary experts to ensure fairness and reliability.

Check your Progress 2

1. Sensor fusion integrates data from LiDAR, radar, and cameras to create an accurate and reliable perception of the environment in autonomous vehicles. It enhances object detection, distance measurement, and decision-making by combining complementary strengths and ensuring redundancy. However, limitations include high computational demands, sensor calibration issues, weather sensitivity, cost constraints, and difficulties handling rare or unpredictable road scenarios.
2. Sentiment analysis helps AI-based customer services to understand the emotions of the customer and the feedback which allows them to respond more accurately and improve user satisfaction.
3. Relying heavily on AI for customer complaint resolution may reduce human empathy and personalization, leading to customer dissatisfaction. AI systems may misinterpret complex queries, sarcasm, or emotional tone, resulting in inappropriate responses. Data privacy concerns, algorithmic bias, system errors, and over-automation can further damage trust, especially when critical or sensitive issues require human judgment and discretion.

Check your Progress 3

1. AI-based adaptive learning systems improve student performance by delivering personalized instruction tailored to each learner's needs, abilities, and pace. These systems continuously analyze student performance data to identify strengths, weaknesses, and knowledge gaps. Based on this analysis, they adjust the difficulty level of content, recommend targeted practice, and provide immediate feedback. This real-time support helps students correct mistakes quickly and reinforces conceptual understanding. Additionally, adaptive systems promote self-paced learning, reducing frustration and improving confidence. By enhancing engagement through interactive and data-driven approaches, AI-based adaptive learning systems increase knowledge retention, academic achievement, and overall learning effectiveness compared to traditional methods.
2. Over-reliance on AI tools may weaken students' critical thinking and problem-solving skills by reducing active cognitive engagement. When learners depend heavily on AI-generated answers, they may focus more on obtaining solutions than understanding underlying concepts. This can limit analytical reasoning, creativity, and independent judgment. Additionally, constant assistance may discourage perseverance in tackling complex problems. Students might also struggle to evaluate the accuracy or bias of AI outputs, affecting their ability to think critically.
3. In AI-supported classrooms, teachers should evolve into facilitators, mentors, and ethical guides. Rather than merely delivering content, they

must interpret AI-generated insights, foster critical thinking, and ensure responsible technology use. Teachers should use learning analytics to personalize instruction while providing emotional support, creativity, and human interaction that AI cannot replace, ensuring balanced and meaningful learning experiences.

12.9 FURTHER READINGS

- S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Hoboken, NJ, USA: Pearson Education, 2021.
- E. Rich, K. Knight, and S. B. Nair, *Artificial Intelligence*, 3rd ed. New Delhi, India: McGraw Hill Education, 2017.
- S. Chopra, *Artificial Intelligence and Machine Learning: Fundamentals and Applications*. New Delhi, India: Khanna Publishing, 2022.
- S. Jain, K. Pandey, P. Jain, and K. P. Seng, *Artificial Intelligence, Machine Learning, and Mental Health in Pandemics: A Computational Approach*, 1st ed. Academic Press, 2022. ISBN 9780323911962.
- P. J. Rousseeuw, *Artificial Intelligence and Big Data in Financial Analytics*. Cham, Switzerland: Springer, 2022.
- R. Baker and K. Yacef, *The State of Educational Data Mining in 2020: AI in Learning and Teaching*. New York, NY, USA: Springer, 2020.

12.11 REFERENCES

- [1] Jie, Zhou, Zeng Zhiying, and Li Li. "A meta-analysis of Watson for Oncology in clinical application." *Scientific reports* 11.1 (2021): 5792.
- [2] <https://d3.harvard.edu/platform-digit/submission/robo-banking-artificial-intelligence-at-jpmorgan-chase/>
- [3] <https://www.tesla.com/AI>
- [4] <https://www.britannica.com/technology/Amazon-Alexa>
- [5] <https://www.duolingo.com/>